

Received March 21, 2020, accepted March 28, 2020, date of publication April 7, 2020, date of current version April 22, 2020.

Digital Object Identifier 10.1109/ACCESS.2020.2986389

A Comparative Study on Arrhenius Equations and BP Neural Network Models to Predict Hot Deformation Behaviors of a Hypereutectoid Steel

LING QIAO¹, YONG DENG², YUAN WANG³, AND JINGCHUAN ZHU¹

¹School of Material Science and Engineering, Harbin Institute of Technology, Harbin 150001, China

²State Key Laboratory for Advanced Metals and Materials, University of Science and Technology Beijing, Beijing 10083, China

³College of Information Science and Engineering, Northeastern University, Shenyang 110819, China

Corresponding author: Jingchuan Zhu (pg_hgd@163.com)

ABSTRACT In order to predict the high-temperature deformation behavior of hypereutectoid steel, the hot compression tests were conducted in the strain rate range of $(0.001 \sim 1) \text{ s}^{-1}$ and the deformation temperature range of $(950 \sim 1100)^\circ\text{C}$. The experimental data were employed to develop the Arrhenius constitutive model and BP neural network model, and their predictability for high temperature flow stress of hypereutectoid steel was further evaluated. Comparatively, a higher correlation coefficient (R) can be obtained for the BP neural network model compared with the Arrhenius constitutive equations. And the relative error within $\pm 1\%$ was more than 68.75% for the BP neural network model, while only 18.75% for the constitutive equations. The BP neural network model is considered more efficient and accurate to predict the hot deformation behavior than the Arrhenius constitutive equations. Moreover, the well-trained BP neural network model is employed to predict the flow stress varying with the deformation temperature and strain rate. The flow stress decreases with the increasing deformation temperature and decreasing strain rate, which is in accordance with the experimental evaluation. The results indicate that BP neural network model is an efficient tool for modelling and predicting the flow behavior of hypereutectoid steels in high temperature applications.

INDEX TERMS Hypereutectoid steel, hot deformation behavior, Arrhenius equations, BP neural network models.

I. INTRODUCTION

Hypereutectoid steel is widely applied in many industrial fields for its great mechanical property, especially in the field of rail technology [1]–[3]. The understanding of the flow behaviors of steels has become more essential to researchers engaged in hot deformation condition. To improve the mechanical performance of hypereutectoid steels, an exploration of the deformation behavior at high temperature has become a critical issue.

Knowledge of the hot deformation mechanisms is considered to be an essential prerequisite for the optimization of material properties [4]. However, the researches on the hypereutectoid steel is rare and this particular subject needs to be

investigated in great depth. Generally, the constitutive model is considered to be a great analytic method to describe the hot deformation behavior of materials based on the experimental data. With the great progress in the development of constitutive relationship, it primarily consists of the phenomenological, physically based and artificial neural network models [5]. Without special considerations on the essential physical process, the Arrhenius type model was successfully used to describe the relationships between the mechanical properties and hot working conditions mathematically [6]–[8]. In particular, Wu *et al* [9] proposed that the constitutive equation can be established to understand the thermal deformation mechanism and thus optimizing the thermal processing parameters and leading to improved thermal processing properties of steels. Zhou *et al.* [10] established the constitutive equation to predict the value of peak stress of Ni3Al-based alloy in

The associate editor coordinating the review of this manuscript and approving it for publication was Kathiravan Srinivasan¹.

hot deformation and gave the processing maps to determine the optimal processing parameters and unstable domains for hot working. Samantaray *et al.* [11] employed the Johnson Cook model, Zerilli-Armstrong model, and the Arrhenius-type constitutive equations to describe the flow behavior of modified 9Cr-1Mo steel respectively and the comparison presented that the Arrhenius-type equations exhibit better performance than the others.

In recent year, the employment of artificial neural network (ANN) method in the prediction of constitutive relationships received considerable attention, especially to solve highly nonlinear and complex problems [12]–[27]. Lin *et al.* [5], [6], [12], [16]–[18], Chen [19] employed phenomenological constitutive models, deep belief network model and adaptive-network-based fuzzy inference system model to describe the flow behaviors of various materials and showing a very good correlation. These works can verify the reliability of neural network. Duan [23] employed the Arrhenius-type model, multiple-linear regressed model and artificial neural network (ANN) model to describe the hot deformation behavior and microstructural evolution of Pb-Mg-10Al-0.5B alloy. And this work demonstrates the higher accuracy of ANN model. Since the artificial neural network learns from the experimental data and identify the relevant parameters and their interactions in a series of input and output values without material constants, it has become a powerful tool to establish the constitutive relationship and further predict the flow behaviors during hot compression process.

In present paper, hot compression tests were performed to obtain the rheological curves. Subsequently, constitutive relationships based on the Arrhenius constitutive equation and a three-layer feed-forward ANN model are employed to characterize the hot deformation behavior of the hypereutectoid steels. And the suitability of these two models to predict the hot deformation behavior was evaluated. Finally, the coupled effects of the deformation temperature and strain rate on the flow behavior were systematically investigated and provided a reference for the actual industrial application.

II. EXPERIMENTAL PROCEDURE

The material used in this study was hypereutectoid steel and the chemical composition (wt%) was plotted in Table.1. Then these materials were processed into cylindrical specimens with a diameter of 6 mm and a height of 9 mm. The hot compression tests were carried out on a Gleeble-1500D thermo-mechanical simulator in the strain rate of 1 s^{-1} , 0.1 s^{-1} , 0.01 s^{-1} , 0.001 s^{-1} and the deformation temperature of 950°C , 1000°C , 1050°C , 1100°C , respectively. To minimize the deformed friction between the specimens and simulator, the graphite was used as lubricant mixed with machine oil. Each sample was heated to the deformation temperature at a rate of 10 K/s and the samples are homogenized for 3 min before compression. During the examination, a thermocouple at the center of specimen was used to monitor the temperature while the stress-strain data at different temperatures

TABLE 1. Chemical composition (wt%).

C	Si	Mn	Cr	Cu	Ni	Nb	V	P	S
0.95	0.50	0.91	0.20	0.04	0.02	0.002	0.06	0.009	0.004

and strain rates were recorded automatically. Furthermore, the deformation in height was 75% at the end of compression tests and the specimens were consequently taken out and then immediately water quenched to room temperature.

III. RESULTS AND DISCUSSIONS

A. FLOW BEHAVIOR ANALYSIS

Fig.1 depicts the typical true stress-strain curves obtained from the hot compression tests of the hypereutectoid steel. Apparently, the evolution of all the flow stress curves follows the same trend. With the work hardening at the initial stage of deformation, the flow stresses are increasing rapidly with the increase of strain and reach a peak value [28]. Attributed to the dynamic recrystallization, the softening effect contributes to a slight decrease in the flow stress. Until arriving at a balance of the work hardening and flow softening, subsequently, a steady state was drawn for the flow stress at the higher strain. Accordingly, the flow stress tends to be intensely dependent on the forming temperature and strain rate under the working conditions. It is found that the flow stress decreases with the increase in deformation temperatures for a given strain rate. The increasing temperature would decrease the critical shear stress of each slip system and increase the thermal activation energy, thus contributing to the dynamic recovery recrystallization. While for a specific temperature, the flow stress generally increases with the increase in strain rates due to an increase of dislocation density and the dislocation multiplication rate. Moreover, the increasing strain rate tends to result in the inhomogeneity of plastic deformation and there is no enough time for the necessary deformation [29].

B. ARRHENIUS CONSTITUTIVE EQUATION

To further investigate the deformation behaviors of hypereutectoid steels, the constitutive equation is developed based on the experimental results. In the present work, Arrhenius equation is used to describe the flow behavior at high temperature, specifically the relationships between flow stress, the strain rate, and deformation temperature. For the low stress level and high stress level, the relationships between the flow stress and strain rate can be expressed in the following equations, respectively.

$$\dot{\epsilon} = A_1 \sigma^{n_1} \exp\left(\frac{-Q}{RT}\right) \quad (1)$$

$$\dot{\epsilon} = A_2 \exp(\beta\sigma) \exp\left(\frac{-Q}{RT}\right) \quad (2)$$

For all the stress level, the flow stress and strain rate meet the hyperbolic sine function and can be represented as

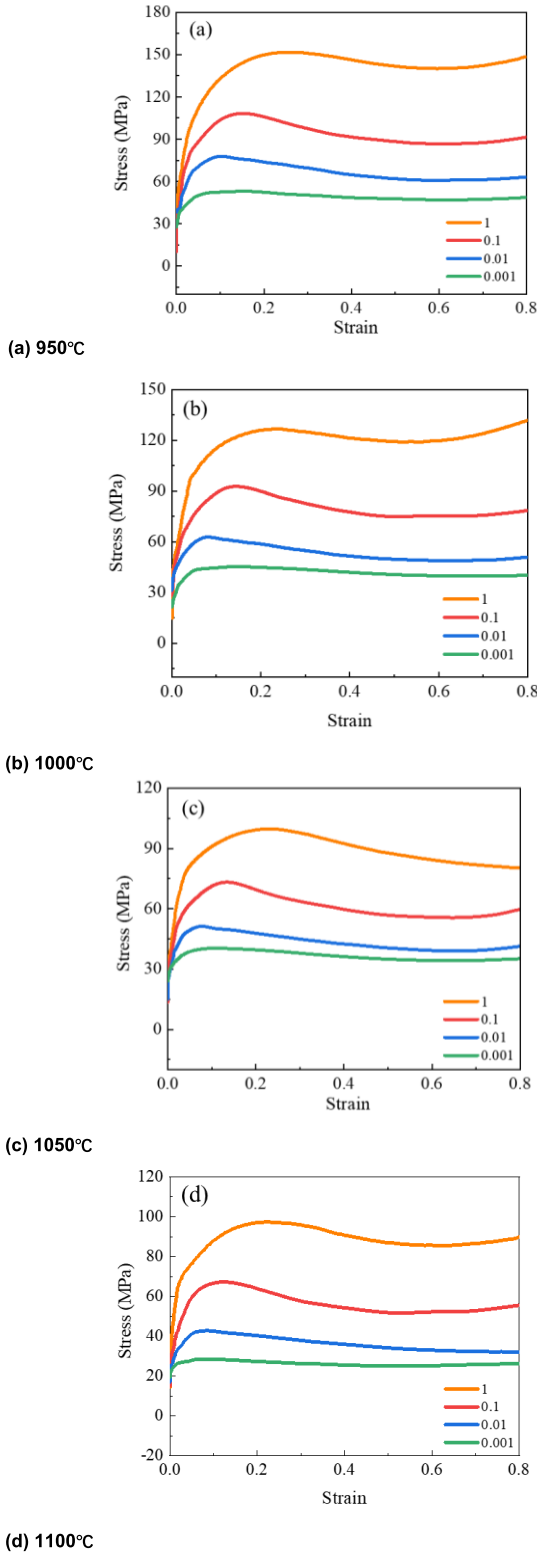


FIGURE 1. True stress-strain curves under different forming temperature and strain rates.

the equation (3).

$$\dot{\epsilon} = A_3 [\sinh(\alpha\sigma)]^n \exp\left(\frac{-Q}{RT}\right) \quad (3)$$

where σ is the flow stress, $A_1, A_2, A_3, n_1, \alpha, \beta$ and n are the material constants. In addition, Zener-Hollomon parameter Z can be used to estimate the effects of deformation temperature and strain rate on the deformation behaviors, which is expressed in the exponential equation (4) [30].

$$Z = \dot{\epsilon} \exp\left(\frac{Q}{RT}\right) = A_3 [\sinh(\alpha\sigma)]^n \quad (4)$$

where $\dot{\epsilon}$ is strain rate (s^{-1}), T is the deformation temperature (K), R is universal gas constant ($R = 8.3154 \text{ J/K/mol}$) and Q is the activation energy ($\text{kJ}\cdot\text{mol}^{-1}$). Take natural logarithm at both sides at equations (1) and (2), respectively, and the equations can be expressed as equations (5) and (6).

$$\ln \dot{\epsilon} = \ln A_1 - \frac{Q}{RT} + n_1 \ln \sigma \quad (5)$$

$$\ln \dot{\epsilon} = \ln A_2 - \frac{Q}{RT} + \beta \sigma \quad (6)$$

At a given deformation temperature, the partial differential equations of equations (5) and (6) can be obtained as follows.

$$n_1 = \left[\frac{\partial \ln \dot{\epsilon}}{\partial \ln \sigma} \right]_T \quad (7)$$

$$\beta = \left[\frac{\partial \ln \dot{\epsilon}}{\partial \sigma} \right]_T \quad (8)$$

Take the linear regression and the Fig.2 gave a detailed description on the $\ln \dot{\epsilon} - \ln \sigma$ and $\ln \dot{\epsilon} - \sigma$ at different temperatures. The values of n_1 and β are determined as the slopes of the curves. Thus the average value can be obtained as $n = 6.56 \text{ MPa}^{-1}$ and $\beta = 0.09 \text{ MPa}^{-1}$. Then the value of α can be obtained according to $\alpha = \beta/n_1 = 0.0137 \text{ MPa}^{-1}$.

Take natural logarithm at both sides at equations (3), it can be expressed as follows.

$$\ln \dot{\epsilon} = \ln A_3 + n \ln [\sinh(\alpha\sigma)] - \left(\frac{Q}{RT} \right) \quad (9)$$

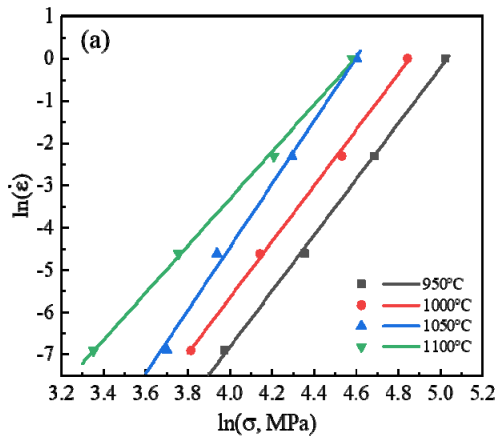
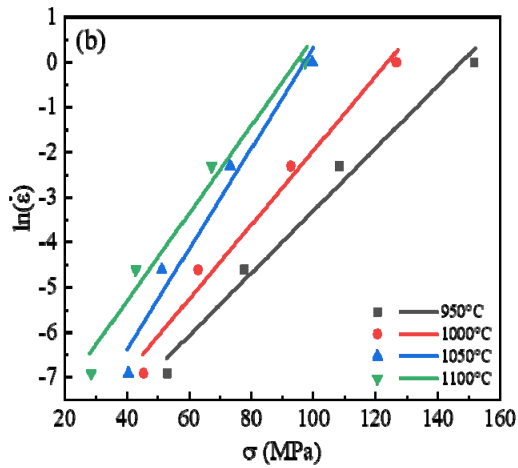
To obtain the plastic deformation behavior at high temperature, the activation energy is assumed to be stabilized in a small temperature range when the strain rate remains constant. Then, take partial differential to equation (9), thus n and Q can be expressed as equation (10) and (11), respectively.

$$n = \left[\frac{\partial \ln \dot{\epsilon}}{\partial \ln [\sinh(\alpha\sigma)]} \right]_T \quad (10)$$

$$Q = R \left[\frac{\partial \ln \dot{\epsilon}}{\partial \ln [\sinh(\alpha\sigma)]} \right]_T \left[\frac{\partial \ln [\sinh(\alpha\sigma)]}{\partial (1/T)} \right]_{\dot{\epsilon}} \quad (11)$$

Subsequently, the linear regression analysis is performed and the $\ln \dot{\epsilon} - \ln [\sinh(\alpha\sigma)]$ and $\ln [\sinh(\alpha\sigma)] - 1/T$ at different strain rates were described in Fig.3. The averaged value of n and b can be obtained as $n = 4.890$, $b = 8224.0$, respectively, thus the average activation energy can be derived from the slopes of $\ln [\sinh(\alpha\sigma)] - 1/T$ plots as $Q = Rnb = 334.4 \text{ kJ/mol}$.

Take logarithmic function at both side of equation (4), thus the equation (12) can be obtained. Then take linear regression fitting on the data points of $\ln Z$ and $\ln [\sinh(\alpha\sigma)]$ additionally,

(a) $\ln \dot{\epsilon} - \ln \sigma$ (b) $\ln \dot{\epsilon} - \sigma$ **FIGURE 2.** Diagram of strain rate and flow stress at different temperatures.

the averaged fitting line is given in Fig.4. As can be seen, most of the data points lie very close to the line and the correlation coefficients (R) is determined as 0.993.

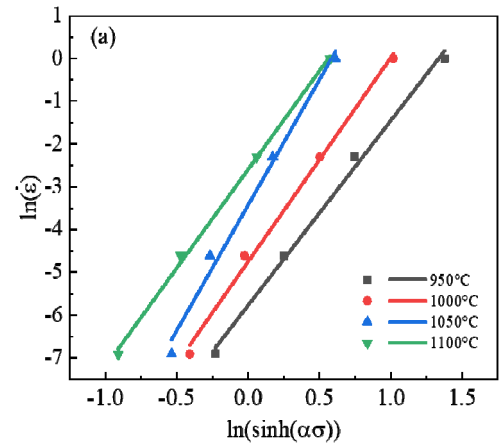
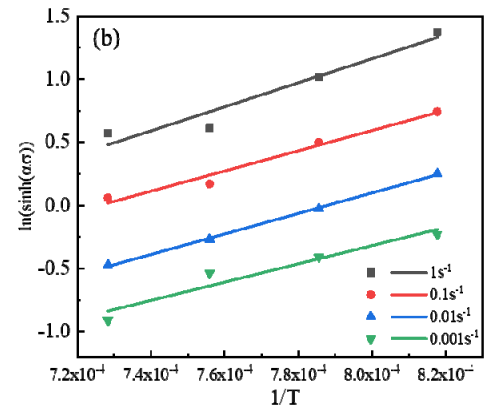
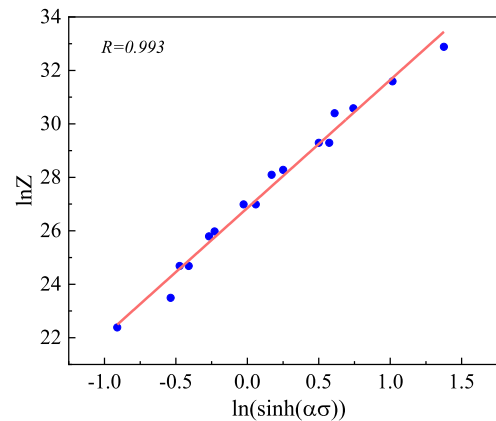
$$\ln Z = \ln A_3 + n \ln[\sinh(\alpha\sigma)] \quad (12)$$

A detailed analysis of the deformation constitutive equations at high temperature is necessary in process of compression deformation. Specifically, regression analysis gives $n = 4.791$ and $\ln A_3 = 26.853$. Based on the above analysis, the mathematic model of the hypereutectoid steel can be determined as the equation (13). Consequently, it is beneficial to describe the flow behavior and provide theory basis for hot deformation analysis.

$$\ln \dot{\epsilon} = 26.853 + 4.791 \times \ln[\sinh(0.0137\sigma)] - \frac{40215}{T} \quad (13)$$

C. ARTIFICIAL NEURAL MODEL

In recent years, the excellent self-adaptive and self-learning ability enable the BP neural network an effective analytical tool to model the hot deformation behavior of metallic materials [31]. The BP neural network model includes

(a) $\ln \dot{\epsilon} - \ln[\sinh(\alpha\sigma)]$ (b) $\ln[\sinh(\alpha\sigma)] - 1/T$ **FIGURE 3.** Diagram of strain rate and temperature at different strain rates.**FIGURE 4.** Diagram of $\ln Z - \ln[\sinh(\alpha\sigma)]$.

input layer, hidden layer and output layer [32], [33]. So far, it tends to be especially useful to address problems where the relationships between inputs and outputs are not sufficiently known. It shows an excellent ability to learn considerable data without any assumptions about their interrelations, and then is used to predict unknown values [34]. In this paper, a multi-layered BP neural network model was constructed on the basis of experimental data obtained from the investigation

on the hot working behavior of the steel. The flow stress is considered to be estimated, while the deformation temperature and strain rate were used as input data.

To obtain the great generalization ability and avoid over-fitting phenomenon, Bayesian regularization algorithm was applied to improve the BP neural network. Remarkably, the neurons in the multilayer BP neural network are considered to be fully interconnected by weights, and the optimal number of hidden layers and the neurons in each layer tends to exhibit significant effects. During this examination, the neurons in the input layer denote the hot working parameters, including the deformation temperature and lg strain rate, and the neurons in the output layer denote the flow stress. To find the optimal architecture, the architecture containing neurons in the hidden layer are investigated and the accuracy of the predicted values is monitored before the optimal structure is determined. And the weights and thresholds of networks are constantly revised until acquiring the expected output. Finally, the number of nodes in the hidden layer is determined to 7, which can avoid over-fitting and simultaneously improve the computing efficiency. In this work, the learning rate was determined to be 0.05 with the allowed target error of 0.00005, and the maximum number of learning was set to 3000.

Specifically, the mean squared error (MSE) is used for comparison between the experimental and predicted values to assess the quality of the proposed neural network model. The criterion of *MSE* can be expressed as follows [35].

$$MSE = \frac{1}{n} \sum_{i=1}^n (A_i - O_i)^2 \quad (14)$$

where, A_i and O_i are the actual output and desired output at the output neuron i , respectively. The *MSE* is monitored during training process until no significant improvement exists in the error. The training performance is illustrated in Fig.5. The asymptotes of flow stress values were obtained from the BP neural networks and the experimental values. In general, the regression value (*R*) is of such importance and determines the correlation between the outputs and targets. Knowing that a great training performance can be confirmed with the *R* values near to 1. In this case, despite a little scatter around line of equality between the analytical and predicted values of flow stress, the high regression value of 0.995 suggests a great fitting effect. Thus, the BP neural networks developed in this work can learn for the training data and incorporate it into its strategy to achieve perfect accuracy.

D. COMPARISON OF THE ARRHENIUS EQUATIONS AND THE BP NEURAL NETWORK MODEL

The flow stress varying with temperature and strain rate was predicted through the Arrhenius constitutive equation and the BP neural network model in this work. Thus then, the variation between experimental data and the predicted data is evaluated. As can be seen from Fig.5, the consensus between the experimental values and the predicted ones

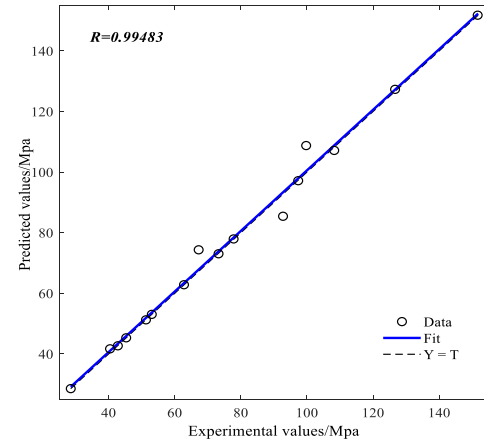


FIGURE 5. Training results.

based on the BP neural network model is very good for all tested temperature and strain rate range. However, there is a remarkable deviation between experimental values and the predicted values from the Arrhenius constitutive equations under some conditions, such as at 1050°C in 0.1 s⁻¹ and 1100°C in 0.1 s⁻¹ respectively.

Furthermore, the accuracy of the Arrhenius equation and the BP neural network model was further quantified by the statistical analysis of correlation coefficient (*R*) and relative errors (η). As the Fig.4 and Fig.5 illustrate, the values of *R* for the constitutive equations and the BP neural network model are 0.993 and 0.995, respectively, which suggests that the developed BP neural network model has more excellent predictability than the constitutive model. Additionally, the relative errors (η) between the experimental values and predicted values were determined as follows.

$$\eta(\%) = \frac{P_i - E_i}{E_i} \times 100\% \quad (15)$$

where E represents the experimental value and P represents the predicted value by one model. The relative errors (η) of Arrhenius constitutive equations and BP neural network model were depicted as the relative frequency versus relative error plot. As illustrated in Fig.6, the relative errors obtained from the constitutive equations are in the range of -8.6%~13.9%; whereas those are found to vary from -10.5% to 7.9% for the BP neural network model. Meanwhile, the relative error (η) within $\pm 1\%$ was observed for more than 68.75% of predicted data set of the BP neural network model, while only 18.75% of the predicted data sets for the constitutive equations. Obviously, significant distinctions can be drawn between the two models and the BP neural network model performs more accurately.

Comparatively, the Arrhenius equations obtain a relatively rough estimate for the flow stress. Since the hot deformation behavior not only depends on the hot working conditions, including the deformation and strain rates, but also affected by many other factors. The relationships are very complex and highly non-linear. However, unlike the constitutive methods, the BP neural network model learns from experimental

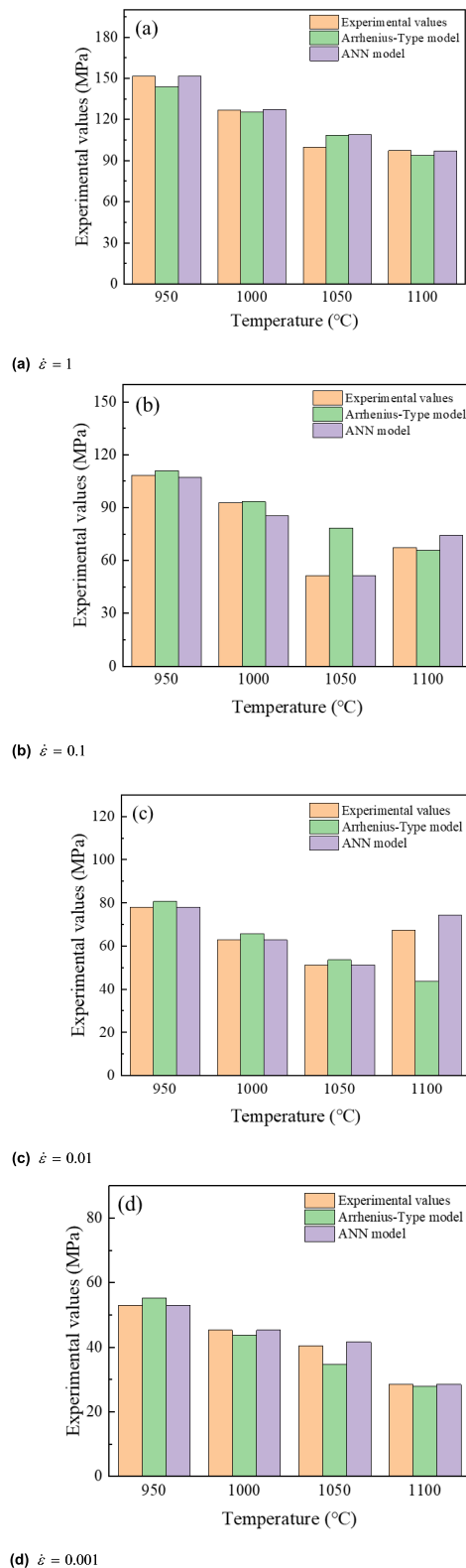


FIGURE 6. Comparison of experimental values and predicted values by Arrhenius constitutive equations and the BP neural network model.

data and recognizes its interrelations instead of a mathematical equation without considering the material constant. Thus, the results indicate that the BP neural network model

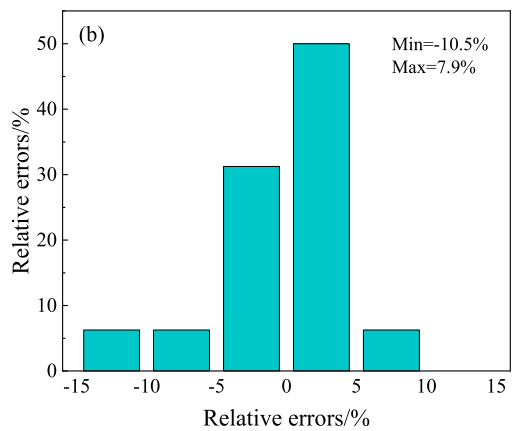
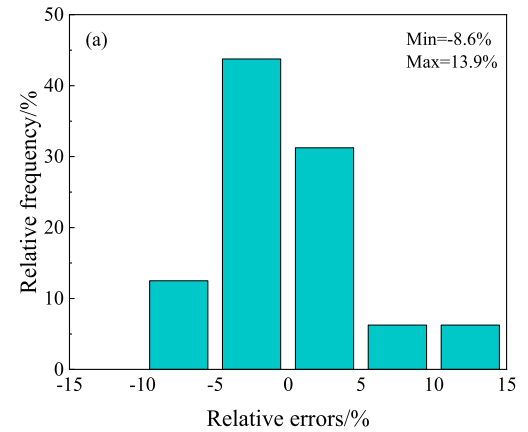


FIGURE 7. Distributions on prediction error by models.

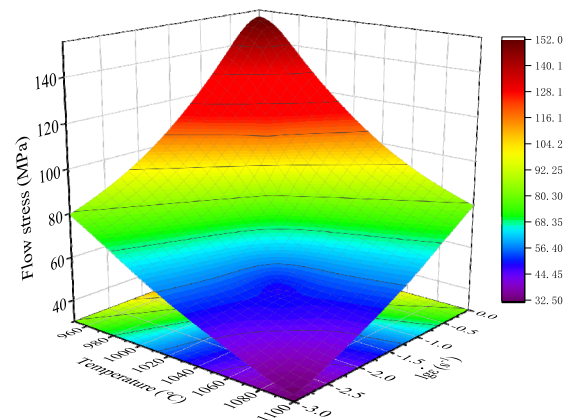


FIGURE 8. Effect of temperature and strain rate on the flow stress.

performs a highly accurate non-linear fit and is particularly more efficient than the constitutive equations to predict the flow behavior of hypereutectoid steel.

E. PREDICTION OF THE FLOW BEHAVIOR

The BP neural network model established above shows a great performance after training, thus can be used with reasonable precision to extend the existing predictions beyond

the availability of experimental data. To give a more detailed understanding, the flow stress varying with deformation temperature and strain rate was predicted through the developed BP neural network model in this work. Thus, a counter plot of the flow stress was drawn on the three-dimensional plane of deformation temperature and strain rate, as illustrated in Fig.8. The data points on the surface plots are adopted from the results predicted by the BP neural network model and most of the data points were outside the experimental conditions. The results suggest that the flow stress varies from 32.8MPa to 151.8MPa with the decrease in temperature from 950°C to 1100°C and the increase in the strain rate from 0.01 to 10 s⁻¹. In particular, the variation obtained from the predicted value by BP neural network model agrees well with the experimental evaluation, which indicates that the proposed BP neural network model can be used to predict the flow stress behavior instead of a considerable amount of experimental work.

IV. CONCLUSION

The present study analyzed the hot deformation behavior of hypereutectoid steel, investigated by a series of thermal deformation tests in the strain rate range of (0.001~1) s⁻¹ and the deformation temperature range of (950~1100) C°. The experimental stress-strain data were employed to develop the Arrhenius constitutive equations and the BP neural network model. And a comparative investigation was carried out on their predictability of the hot deformation behavior of hypereutectoid steel. The conclusions can be drawn as follows.

(1) The predictability of the Arrhenius constitutive equations and BP neural network model was further evaluated in terms of the correlation coefficient (R) and the relative error(η). For the Arrhenius constitutive equations, R value is determined as 0.993, while for the BP neural network model it is found to be 0.995, respectively. The relative errors between experimental and predicted flow stress obtained from the constitutive equations and BP neural network model are in the range of -8.6%~13.9% and -10.5% to 7.9%, respectively. Moreover, the relative error within $\pm 1\%$ was observed for more than 68.75% for the BP neural network model, while only 18.75% for the constitutive equations. Thus, the developed BP neural network model has better capability compared with the Arrhenius constitutive equations.

(2) The BP neural network model for hypereutectoid steel can accurately predict the flow stress behavior across a wider range of temperatures and strain rates. The coupled effects of the deformation temperature and strain rate on the flow behavior were systematically investigated and the evolution indicates that the flow stress decreases with the increasing deformation temperature and decreasing strain rate. Thus, the developed BP neural network model could not only quite well predict the hot deformation behavior but also significantly reduce time and cost for the analyses.

REFERENCES

- [1] P. Boonsukachote, S. Kingklang, and V. Uthaisangsk, "Modelling of mechanical properties of pearlitic rail steel," *Key Eng. Mater.*, vol. 798, pp. 3–8, Apr. 2019.
- [2] R. Masoudi Nejad, K. Farhangdoost, and M. Shariati, "Microstructural analysis and fatigue fracture behavior of rail steel," *Mech. Adv. Mater. Struct.*, vol. 27, no. 2, pp. 152–164, Jan. 2020.
- [3] Y. Lan, G. Zhao, Y. Xu, C. Ye, and S. Zhang, "Effects of quenching temperature and cooling rate on the microstructure and mechanical properties of U75 V rail steel," *Metallography, Microstruct., Anal.*, vol. 8, no. 2, pp. 249–255, Apr. 2019.
- [4] S. Spigarelli, M. E. Mehtedi, M. Cabibbo, F. Gabrielli, and D. Ciccirelli, "High temperature processing of brass: Constitutive analysis of hot working of Cu–Zn alloys," *Mater. Sci. Eng., A*, vol. 615, pp. 331–339, Oct. 2014.
- [5] Y. C. Lin and X.-M. Chen, "A critical review of experimental results and constitutive descriptions for metals and alloys in hot working," *Mater. Des.*, vol. 32, no. 4, pp. 1733–1759, Apr. 2011.
- [6] Y. C. Lin, L.-T. Li, and Y.-Q. Jiang, "A phenomenological constitutive model for describing thermo-viscoplastic behavior of Al–Zn–Mg–Cu alloy under hot working condition," *Experim. Mech.*, vol. 52, no. 8, pp. 993–1002, Oct. 2012.
- [7] J. E. Plumeri, A. Madej, and W. Z. Misiolok, "Constitutive modeling and inverse analysis of the flow stress evolution during high temperature compression of a new ZE20 magnesium alloy for extrusion applications," *Mater. Sci. Eng., A*, vols. 740–741, pp. 174–181, Jan. 2019.
- [8] N. R. Jaladurgam and A. K. Kanjarla, "Hot deformation characteristics and microstructure evolution of hastelloy C-276," *Mater. Sci. Eng., A*, vol. 712, pp. 240–254, Jan. 2018.
- [9] Y. Wu, Y. Liu, C. Li, X. Xia, Y. Huang, H. Li, and H. Wang, "Deformation behavior and processing maps of ni 3 al-based superalloy during isothermal hot compression," *J. Alloys Compounds*, vol. 712, pp. 687–695, Jul. 2017.
- [10] Y. Zhou, Y. Liu, X. Zhou, C. Liu, J. Yu, Y. Huang, H. Li, and W. Li, "Precipitation and hot deformation behavior of austenitic heat-resistant steels: A review," *J. Mater. Sci. Technol.*, vol. 33, no. 12, pp. 1448–1456, Dec. 2017.
- [11] D. Samantaray, S. Mandal, and A. K. Bhaduri, "A comparative study on Johnson cook, modified zerilli–armstrong and arrhenius-type constitutive models to predict elevated temperature flow behaviour in modified 9Cr–1Mo steel," *Comput. Mater. Sci.*, vol. 47, no. 2, pp. 568–576, Dec. 2009.
- [12] Y. C. Lin, M.-S. Chen, and J. Zhong, "Constitutive modeling for elevated temperature flow behavior of 42CrMo steel," *Comput. Mater. Sci.*, vol. 42, no. 3, pp. 470–477, May 2008.
- [13] M. Arjomandi, S. H. Sadati, H. Khorsand, and H. Abdoos, "Austenite formation temperature prediction in steels using an artificial neural network," *Defect Diffusion Forum*, vols. 273–276, pp. 335–341, Feb. 2008.
- [14] H.-Y. Li, D.-D. Wei, Y.-H. Li, and X.-F. Wang, "Application of artificial neural network and constitutive equations to describe the hot compressive behavior of 28CrMnMoV steel," *Mater. Design*, vol. 35, pp. 557–562, Mar. 2012.
- [15] S. Mandal, P. V. Sivaprasad, and S. Venugopal, "Constitutive flow behaviour of austenitic stainless steels under hot deformation: Artificial neural network modelling to understand, evaluate and predict," *Model. Simul. Mater. Sci. Eng.*, vol. 14, no. 6, pp. 1053–1070, 2006.
- [16] Y. C. Lin, J. Zhang, and J. Zhong, "Application of neural networks to predict the elevated temperature flow behavior of a low alloy steel," *Comput. Mater. Sci.*, vol. 43, no. 4, pp. 752–758, Oct. 2008.
- [17] Y. C. Lin, J. Li, M.-S. Chen, Y.-X. Liu, and Y.-J. Liang, "A deep belief network to predict the hot deformation behavior of a ni-based superalloy," *Neural Comput. Appl.*, vol. 29, no. 11, pp. 1015–1023, Jun. 2018.
- [18] Y. C. Lin, Y.-J. Liang, M.-S. Chen, and X.-M. Chen, "A comparative study on phenomenon and deep belief network models for hot deformation behavior of an Al–Zn–Mg–Cu alloy," *Appl. Phys. A*, vol. 123, no. 1, Jan. 2017.
- [19] D.-D. Chen, Y. C. Lin, Y. Zhou, M.-S. Chen, and D.-X. Wen, "Dislocation substructures evolution and an adaptive-network-based fuzzy inference system model for constitutive behavior of a ni-based superalloy during hot deformation," *J. Alloys Compounds*, vol. 708, pp. 938–946, Jun. 2017.
- [20] G. Ji, F. Li, Q. Li, H. Li, and Z. Li, "A comparative study on arrhenius-type constitutive model and artificial neural network model to predict high-temperature deformation behaviour in Aermet100 steel," *Mater. Sci. Eng., A*, vol. 528, nos. 13–14, pp. 4774–4782, May 2011.

- [21] R. Bobbili, B. Venkata Ramudu, and V. Madhu, "A physically-based constitutive model for hot deformation of Ti-10-2-3 alloy," *J. Alloys Compounds*, vol. 696, pp. 295–303, Mar. 2017.
- [22] Z. Zhu, Y. Lu, Q. Xie, D. Li, and N. Gao, "Mechanical properties and dynamic constitutive model of 42CrMo steel," *Mater. Des.*, vol. 119, pp. 171–179, Apr. 2017.
- [23] Y. Duan, L. Ma, H. Qi, R. Li, and P. Li, "Developed constitutive models, processing maps and microstructural evolution of Pb-Mg-10Al-0.5B alloy," *Mater. Characterization*, vol. 129, pp. 353–366, Jul. 2017.
- [24] Y. Duan, P. Li, L. Ma, and R. Li, "Dynamic recrystallization and processing map of Pb-30Mg-9Al-1B alloy during hot compression," *Metall. Mater. Trans. A*, vol. 48, no. 7, pp. 3419–3431, Jul. 2017.
- [25] Z. Cai, H. Ji, W. Pei, X. Tang, X. Huang, and J. Liu, "Hot workability, constitutive model and processing map of 3Cr23Ni8Mn3N heat resistant steel," *Vacuum*, vol. 165, pp. 324–336, Jul. 2019.
- [26] M. S. Arun and U. Chakkingal, "A constitutive model to describe high temperature flow behavior of AZ31B magnesium alloy processed by equal-channel angular pressing," *Mater. Sci. Eng., A*, vol. 754, pp. 659–673, Apr. 2019.
- [27] W. Bao, L. Bao, D. Liu, D. Qu, Z. Kong, M. Peng, and Y. Duan, "Constitutive equations, processing maps, and microstructures of Pb-Mg-Al-B-0.4Y alloy under hot compression," *J. Mater. Eng. Perform.*, vol. 29, no. 1, pp. 607–619, Jan. 2020.
- [28] Y. Sun, Z. Wan, L. Hu, and J. Ren, "Characterization of hot processing parameters of powder metallurgy TiAl-based alloy based on the activation energy map and processing map," *Mater. Design*, vol. 86, pp. 922–932, Dec. 2015.
- [29] F. Warchomicka, C. Poletti, and M. Stockinger, "Study of the hot deformation behaviour in Ti-5Al-5Mo-5 V-3Cr-1Zr," *Mater. Sci. Eng., A*, vol. 528, no. 28, pp. 8277–8285, Oct. 2011.
- [30] C. M. Sellars and W. J. McTegart, "On the mechanism of hot deformation," *Acta Metallurgica*, vol. 14, no. 9, pp. 1136–1138, 1966.
- [31] Z. Y. Chen, D. N. Zou, J. H. Yu, and Y. Han, "Artificial neural network approach to predict mechanical properties of 301 austenitic stainless steel," *Mater. Sci. Forum*, vol. 658, pp. 145–148, Jul. 2010.
- [32] W. K. Wong, C. W. M. Yuen, D. D. Fan, L. K. Chan, and E. H. K. Fung, "Stitching defect detection and classification using wavelet transform and BP neural network," *Expert Syst. Appl.*, vol. 36, no. 2, pp. 3845–3856, Mar. 2009.
- [33] Y.-J. Song, B.-Y. Song, Z.-S. Zhang, and Y.-W. Chen, "The satellite downlink replanning problem: A BP neural network and hybrid algorithm approach for IoT Internet connection," *IEEE Access*, vol. 6, pp. 39797–39806, 2018.
- [34] X. Xiao, G. Q. Liu, B. F. Hu, X. Zheng, L. N. Wang, S. J. Chen, and A. Ullah, "A comparative study on arrhenius-type constitutive equations and artificial neural network model to predict high-temperature deformation behaviour in 12Cr3WV steel," *Comput. Mater. Sci.*, vol. 62, pp. 227–234, Sep. 2012.
- [35] S. Hashem and B. Schmeiser, "Improving model accuracy using optimal linear combinations of trained neural networks," *IEEE Trans. Neural Netw.*, vol. 6, no. 3, pp. 792–794, May 1995.



LING QIAO received the B.S. degree in metal material engineering from the Shandong University of Science and Technology, Qingdao, Shandong, in 2016, and the M.S. degree in materials science and engineering from the China University of Petroleum (East China), Qingdao, Shandong, in 2019. She is currently pursuing the Ph.D. degree in materials science and engineering with the Harbin Institute of Technology, Harbin, Heilongjiang, China.

Her research interest includes new material design.



YONG DENG received the B.S. degree in press working from the Xi'an University of Architecture and Technology, Xi'an, China, in 2001, and the M.S. degree in materials science and technology from the University of Science and Technology Beijing, Beijing, China, in 2013, where he is currently pursuing the Ph.D. degree in materials science and technology.

He has been as Special-Grade Researcher with the Ansteel Research Institute of Vanadium and Titanium (iron and steel), Panzhihua, China, since 2001. His research interests include development of rail products and rail processing technologies.

Prof. Deng awards and honors include the successful development of Pangang U71Mn series of heat-treated rails, 60D40 series of high-speed turnout rails, and so on. He has published six articles and 24 patents in the areas of rail steel development and rail processing technology.



YUAN WANG received the B.S. degree in automation from the Shandong University of Science and Technology, Qingdao, Shandong, in 2014, and the M.S. degree in control science and engineering from the China University of Petroleum (East China), Qingdao, Shandong, in 2017. He is currently pursuing the Ph.D. degree in control theory and control engineering with Northeastern University, Liaoning, Shenyang, China.

His research interests include neural networks, and fault-tolerant control and their industrial applications.



JINGCHUAN ZHU received the B.S. degree in metalology and heat treatment from Chongqing University, Chongqing, China, in 1982, and the M.S. and Ph.D. degrees in metal materials and heat treatment from the Harbin Institute of Technology, Harbin, Heilongjiang, China, in 1991 and 1994, respectively.

He is currently a Full Professor with the School of Materials Science and Engineering, Harbin Institute of Technology. He is the author who has more than 100 articles and two books. He has also more than ten patents in the area of materials. His research interest includes multi scale computer simulation and design of materials, functionally gradient composite material design, and so on. He is a Senior Member of China Metal Society, the Deputy Secretary General of Heilongjiang Materials and Testing Society, and the Executive Vice President of Heilongjiang Metal Society.

...